TRPT Design Landscape

Windswept & Interesting Ltd · July 2026 · V10-Spoke · 11 m/s · DE-optimized ring geometry

Background. The TRPT (Tensioned Rotary Power Transmission) kite turbine uses a rotating ring of blades suspended on tensioned tethers to extract wind power at altitude. This report maps the design envelope for the V10-Spoke configuration — 13 viable designs discovered by a 50-point catalog sweep plus targeted kickstart tests, across blade scale (λ 0.69–1.10), generator load (k_{mppt} 2–14), tether diameter (3.5–4 mm), and ring radius (r_{bottom} 1.00–1.30). All simulations use the physically-correct per-vertex Dyneema spoke spring model (tension-only, replacing the earlier centre-constraint model which inflated FoS values), DE-optimized ring tube geometry (D_o=60 mm, t/D=0.01), and 30-second MPPT sustain at 11 m/s wind speed. Designs pass the dual gate of $P \geq 50$ kW ground power and ring Factor of Safety FoS ≥ 1.5 against Euler buckling.

How to read the table. λ (blade scale factor) is swept area relative to ring diameter — larger blades produce more power but increase ring compression. k_{mppt} is the generator load control parameter ($P_{\mathrm{gen}} = k \cdot \omega^3$) — higher k extracts more torque per rpm but lowers FoS. FoS is the ring's buckling safety margin: ≥ 6 is ultra-safe, 4–6 is production-ready, 2.5–4 is adequate with monitoring, < 2.5 is marginal (engineering margin only, needs structural reinforcement for field deployment). Variant superscripts: $^{3.5}$ = 3.5 mm tether, $^{\mathrm{r}}$ = tight ring (r_{bottom}=1.00); unmarked = standard 4 mm tether, r_{bottom}=1.30.

Key findings. Two operating regimes exist: Regime A (high-torque, 168–408 rpm, blades $\geq$ 0.90) and Regime B (high-RPM, 200–482 rpm, blades $\leq$ 0.85). Regime B is partially hidden from the catalog sweep by a settle-to-operational bug (ω ceiling of 91 rpm blinds the optimizer to 300–480 rpm equilibria). Kickstart tests recovered blade 0.85 at k=2 producing 116–167 kW at 326–482 rpm (value varies with start protocol and capture time). The Pareto front runs from blade 0.90·k6 (81 kW, FoS 11.3, safest) through 0.95·k4 (199 kW, FoS 5.2, sweet spot) to 1.10·k4 (297 kW, FoS 2.3, max power). See the companion scatter plot for the full P–ω landscape.

Design	Variant	λ	k	P (kW)	ω (rpm)	FoS	Role
V10·λ0.90·k6	4mm, r1.30	0.90	6	81	168	**11.3**	Safest — ideal for first demonstration
V10·λ0.95·k2$^{\mathrm{r}}$	4mm, r1.00	0.95	2	147	455	**8.6**	Fast, tight ring — high safety at high RPM
V10·λ0.80·k14	4mm, r1.30	0.80	14	133	201	**6.2**	Small blade, high k — compact rotor
V10·λ0.95·k6$^{3.5}$	3.5mm, r1.30	0.95	6	259	301	**6.1**	Light tether, high power, safe
V10·λ1.10·k2$^{\mathrm{r}}$	4mm, r1.00	1.10	2	190	398	**6.1**	Large blade, tight ring — high power, safe
V10·λ0.95·k4	4mm, r1.30	0.95	4	199	257	5.2	⋆ **Sweet spot** — best balance of power & safety
V10·λ1.00·k2$^{\mathrm{r}}$	4mm, r1.00	1.00	2	70	∼420	4.7	Modest power, generous safety margin
V10·λ1.05·k2$^{3.5}$	3.5mm, r1.30	1.05	2	155	318	4.0	Light tether, mid-scale — adequate
V10·λ1.00·k4	4mm, r1.30	1.00	4	146	206	3.9	Balanced baseline design
V10·λ1.05·k2$^{\mathrm{r}}$	4mm, r1.00	1.05	2	196	408	3.5	Tight ring, high RPM — adequate
V10·λ1.00·k2$^{3.5}$	3.5mm, r1.30	1.00	2	223	∼340	2.5	Light tether, power-focused — marginal
V10·λ1.10·k4	4mm, r1.30	1.10	4	297	272	2.3	**Max power** — marginal safety margin
V10·λ1.10·k2$^{3.5}$	3.5mm, r1.30	1.10	2	121	351	2.3	Light tether, large blade — marginal

Variant key: $^{\mathrm{r}}$ = tight ring (r_{bottom}=1.00) $^{3.5}$ = 3.5 mm tether unmarked = standard 4 mm, r_{bottom}=1.30 $\sim\omega$ = estimated from kickstart test (catalog settle-blind)